

Microbiome Power and Effect Size Analysis

Analysis Report

2026-10-07

Setup and Data Loading

First, we load the necessary libraries, set up the project directories, and import the phyloseq object.

Rarefaction and Alpha Diversity Metrics

We rarefy the data to an even depth to standardize sequencing effort across samples, then compute three alpha diversity measures: Observed, Shannon, and Simpson.

Alpha Diversity Power Analysis

Here, we define a function to compute both the p-value and Cohen's f effect size using a one-way ANOVA.

```
=====
Metric: Observed
ANOVA p-value: 0.67073
Estimated effect size (Cohen's f): 0.208
Required total sample size for 80% power: 256
Required sample size per group (assuming equal n): 64
```

```
=====
Metric: Shannon
ANOVA p-value: 0.16783
Estimated effect size (Cohen's f): 0.385
Required total sample size for 80% power: 78
Required sample size per group (assuming equal n): 20
```

```
=====
Metric: Simpson
ANOVA p-value: 0.33316
Estimated effect size (Cohen's f): 0.313
Required total sample size for 80% power: 116
Required sample size per group (assuming equal n): 29
```

Beta Diversity Power Analysis

Next, we calculate the Bray-Curtis and Jaccard distance matrices and run PERMANOVA (`adonis2`). We will extract the exact p-value and R^2 , then convert R^2 to Cohen's f to calculate the required sample sizes.

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Beta Diversity Analysis
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```

Bray-Curtis:

PERMANOVA p-value = 0.871

$R^2 = 0.059$

Jaccard:

PERMANOVA p-value = 0.983

$R^2 = 0.057$

Effect Sizes:

Bray-Curtis $f = 0.251$

Jaccard $f = 0.246$

Bray-Curtis PERMANOVA Power:

Required total N for 80% power: 177

Required N per group: 45

Jaccard PERMANOVA Power:

Required total N for 80% power: 185

Required N per group: 47